

Suraj Dhakad

Backend Developer | AI Engineer

FastAPI • LangGraph • RAG Systems • PostgreSQL • Redis

📞 +91-7879553211 ✉ dhakadsuraj77@gmail.com 🌐 GitHub 🔗 LinkedIn 📁 Portfolio

PROFESSIONAL SUMMARY

Backend Developer and AI Engineer with hands-on experience building production-grade APIs, RAG pipelines, and real-time systems. Proficient in FastAPI, LangGraph, and LangChain for developing LLM-powered applications. Skilled in designing scalable architectures using PostgreSQL, Redis, Docker, and WebSockets. Strong background in AI/ML integration, vector search with FAISS, and multi-agent workflow orchestration.

TECHNICAL SKILLS

- **Languages:** Python, SQL, JavaScript, Bash
- **Backend Frameworks:** FastAPI, Flask, REST APIs, WebSockets, Pydantic, Async Python
- **AI / LLM Stack:** LangChain, LangGraph, LangSmith, RAG Pipelines, Prompt Engineering, Multi-Agent Systems, OpenAI API, Groq API
- **Vector Search & Databases:** FAISS, PostgreSQL, MySQL, Redis, pgvector
- **Authentication & Security:** JWT, OAuth2, RBAC, Secure API Design
- **DevOps & Tools:** Docker, Git, GitHub Actions, Linux, Postman, Railway, Render, Vercel
- **Core Concepts:** System Design, Caching, Authentication, Vector Search, Async Processing

PROJECTS

Perplexity 2.0 — AI-Powered Search Engine | *LangGraph, FastAPI, React, PostgreSQL, FAISS, Redis, SSE* 🌐 GitHub

- Built a production-style AI search engine with **multi-step reasoning, real-time web search, and contextual answer generation** using LangGraph-orchestrated multi-agent workflows.
- Designed a full **RAG pipeline** using FAISS vector store for semantic search and grounded, citation-backed responses — reducing hallucination risk significantly.
- Implemented **Server-Sent Events (SSE)** for streaming responses, achieving low-latency output comparable to production AI systems.
- Integrated **Tavily Search API** for live web retrieval and Redis caching to reduce redundant API calls and improve response throughput.
- Engineered scalable **FastAPI backend** with PostgreSQL for persistent chat memory, user sessions, and source tracking.

Real-Time Chat Application | *FastAPI, WebSockets, Redis Pub/Sub, PostgreSQL, JWT, RBAC* 🌐 GitHub

- Developed a full-stack real-time messaging system using **WebSockets** with support for concurrent users and low-latency delivery.
- Implemented **JWT authentication and Role-Based Access Control (RBAC)** for secure, scoped access across user roles.
- Integrated **Redis Pub/Sub** for horizontal scalability — enabling message broadcasting across multiple server instances.
- Designed **normalized PostgreSQL schema** with modular REST APIs for user management, message history, and room operations.

AI Video Assistant | *Python, OpenAI Whisper, LangChain, RAG, Streamlit, Sarvam AI* 🌐 GitHub

- Built an AI assistant that processes **YouTube videos and local media files** — transcribing, indexing, and enabling conversational Q&A over video content.
- Implemented **speech-to-text transcription** using OpenAI Whisper and a **RAG-based querying pipeline** for contextually accurate answers from video content.
- Generated AI-powered summaries, action items, key decisions, and timestamped highlights automatically using LangChain chains.
- Integrated **Sarvam AI** for multilingual transcription and translation, expanding usability for regional language content.
- Designed a **Streamlit conversational UI** for intuitive real-time interactions with video content — zero prompt-engineering required from end users.

CERTIFICATIONS

- **FastAPI for Machine Learning** — CampusX
- **Complete Agentic AI Bootcamp with LangGraph and LangChain** — Udemy

EDUCATION

B.Tech — Electronics & Telecommunication Engineering

Institute of Engineering & Technology (IET), DAVV Indore

2024 – 2028

Indore, MP